

G Pranav Ganesh

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Summary

Computer Science undergraduate (AI & ML) at PES University with hands-on experience in AI/ML systems, RAG pipelines, and full-stack development. Proven ability to design and deploy end-to-end solutions from fine-tuned transformer models to cloud-based enterprise systems. Actively seeking ML/AI or SDE internship roles.

Experience

Hindustan Aeronautics Limited (HAL)

Jun 2025 – Aug 2025

AI/ML Intern

Bengaluru, India

- Built an **AI-powered semantic search system** using a **RAG pipeline** over large enterprise document corpora, integrating **Google Gemini** for context-grounded answers.
- Engineered a **universal ingestion pipeline** supporting **15+ file formats** (PDF, DOCX, PPTX, CSV, images with OCR, Google Drive) using Sentence-BERT embeddings and **ChromaDB** vector storage.
- Designed prompt strategies to minimize hallucinations; proposed enterprise use cases for knowledge management, codebase search, and automated customer support.

Eduphoenix Solutions

Jan 2024 – Apr 2024

Cybersecurity & Cloud Intern

Bengaluru, India

- Deployed a **cloud-based encryption/decryption system on AWS** using industry-standard algorithms to secure sensitive enterprise data at scale.
- Integrated security solutions into existing cloud infrastructure; deepened expertise in **cybersecurity practices** and scalable system design through hands-on work.

Encode AI PESU

Oct 2025 – Present

Logistics Team Member

Bengaluru, India

- Co-organized a **12-hour hackathon** and a **10-hour competitive coding contest** for **300+ participants**, managing end-to-end logistics: registrations, scheduling, and on-ground coordination.

G Cube PESU

Feb 2025 – Mar 2025

Marketing & Campaigns Team

Bengaluru, India

- Contributed to marketing strategies and campaign execution for a student-led initiative at PES University.

Projects

Financial Sentiment Analysis & Stock Movement Prediction | *Python, RoBERTa, SHAP, yfinance* 2026

- Fine-tuned **RoBERTa** on the FiQA dataset, **outperforming the FinBERT baseline**; engineered novel **disagreement and uncertainty features** to reduce overconfident predictions.
- Applied **SHAP explainability** to audit model decisions; built a downstream **stock movement prediction pipeline** using AAPL and TSLA data via yfinance.
- Documented methodology and experimental results in a complete **IEEE-format research paper**.

XAI Decision Support System | *Python, Conformal Prediction, Flask, Streamlit*

2025–2026

- Extended an **ICML 2024** codebase on counterfactual prediction sets; integrated an FAX Agent pipeline to generate **faithful/faithless XAI metrics** on the ImageNet16H-PS dataset.
- Built a **Flask API** for automated metric analysis and a **Streamlit dashboard** for counterfactual exploration; added 3 new datasets for JMLR-style reproducibility experiments.

Supply Chain Vulnerability Forecasting | *Python, Causal Inference, Network Modeling, ML* 2025

- Linked **climate anomalies** to **sector-level supply chain stress** using causal inference and network modeling; trained Random Forest and XGBoost models to forecast vulnerability scores across supply chain nodes.
- Identified top disruption pathways using **graph-based centrality metrics** and time-series anomaly signals; visualized risk propagation across a multi-sector supply network.

Food Safety Inspection Hub | *Python, Google Gemini AI* [Hackathon 2nd Place, Karnataka State]2024

- Built an anonymous food-safety violation reporting platform with **AI-based validation**, automated authority notifications, and a **Gemini-powered conversational data interface**.
- Won **2nd place** at Craft-N-Code 2024 (Karnataka State), organized by IIIT Bhubaneswar at R.V. College of Engineering.

Education

PES University <i>B.Tech in Computer Science (AI & ML) — Pursuing</i>	Aug 2024 – Aug 2027 <i>Bengaluru, India</i>
Brindavan College of Engineering <i>Diploma in Computer Science CGPA: 8.85 Activities: NCC</i>	Nov 2021 – Jul 2024 <i>Bengaluru, India</i>
Kendriya Vidyalaya <i>10th Standard</i>	Graduated Apr 2021

Skills

AI / ML: RAG Pipelines, Transformer Fine-Tuning (RoBERTa, BERT), SHAP Explainability, Conformal Prediction, CNNs, Sentence-BERT, LLM Prompt Engineering, Google Gemini API, TensorFlow

Cloud & Security: AWS, Docker, ChromaDB, Encryption/Decryption Systems, Cybersecurity Best Practices

Development: Python, JavaScript, Java, SQL, Node.js, React, Flask, Streamlit, MongoDB, MySQL, HTML/CSS, OCR Integration

Tools & Research: Git, GitHub, Jupyter, Postman, Causal Inference, IEEE Paper Writing, Explainable AI (XAI), REST APIs

Certifications & Honors

Fine-Tuning Large Language Models <i>Encode AI PESU</i>	Aug 2025
Deep Learning & CNNs <i>AURA PESU</i>	Nov 2024
Cloud Computing & Cybersecurity <i>Brindavan College of Engineering</i>	
2nd Place — Craft-N-Code Hackathon 2024 (Karnataka State) <i>Organized by IIIT Bhubaneswar, held at R.V. College of Engineering</i>	Oct 2024